

The same encounter is written down **three times**, by three people, and only the last version counts.

USER

Clinicians and health information officers in Nigerian secondary and tertiary facilities.

CURRENT WORKFLOW

1 Clinician writes a free-text consultation note → **2** someone re-keys it into the EMR → **3** someone else codes the diagnosis for reporting and billing → **4** the coded record is what reaches health information systems and funders.

PAIN POINT

The same encounter is manually transcribed and re-interpreted three times. Diagnoses go uncoded, SOAP sections go incomplete, and treatment plans get separated from the diagnoses that justify them.

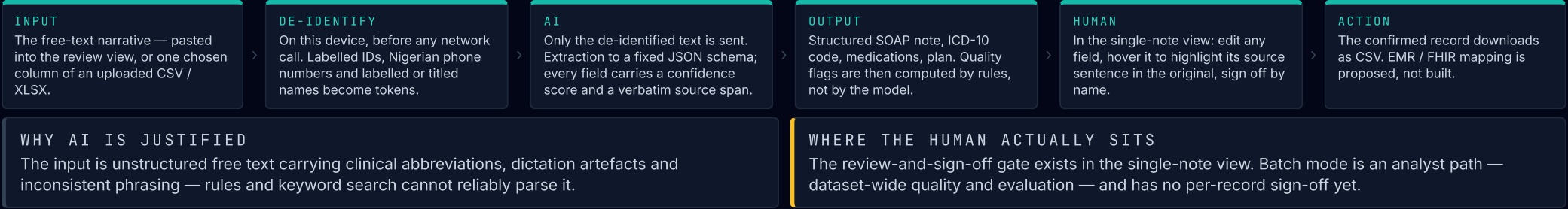
CONSEQUENCE

Incomplete and inconsistent documentation degrades continuity of care, corrupts facility reporting, and surfaces later as data quality findings that put programme funding at risk.

PROBLEM STATEMENT

Clinicians need a better way to convert free-text consultation notes into **complete, coded, structured documentation**, because manual transcription and coding is repeated, inconsistent and unverifiable, resulting in incomplete records and unreliable facility reporting.

MER Capture — one pass from narrative to a coded record the clinician signs.



THE ARCHITECTURAL LINE THAT MATTERS

The LLM only reads and extracts. Every quality flag and every derived metric is computed by deterministic rules over the extracted fields. No reportable output is ever generated by the model.

What is actually built, and what is not.

COMPONENT	STATUS	WHAT IT DEMONSTRATES	WHAT REMAINS
LLM extraction — narrative → fixed JSON schema	WORKING	Per-field confidence and a verbatim source span, located back in the note	A scored run on the organiser dataset; broader schema coverage
Client-side de-identification	PARTIAL	Nigerian phone numbers, labelled hospital / file / patient IDs, slash-delimited codes, and names that are labelled or titled — tokenised before any network call	A bare name in running prose is not caught. Needs a name model or lexicon, and a formal privacy audit
Documentation quality rules — deterministic	WORKING	Six rules: diagnosis without ICD-10, plan without diagnosis, complaint / diagnosis keyword mismatch, medications absent from the plan, incomplete SOAP per core section, low-confidence diagnosis	Clinician-validated rule set
Batch processing across a full dataset	WORKING	Three concurrent extractions, cancellable, one failed row does not stop the run	Throughput and cost at facility volume
Evaluation harness vs reference fields	WORKING	Factual-match scoring, not fluency: exact, token F1, ICD-10 exact and category, hallucination, omission, negation inversion, on a seeded held-out split	A run on the organiser dataset; the non-AI baseline; clinician adjudication
Named clinician sign-off gate	PARTIAL	In the single-note view, export is blocked until a named clinician ticks that they checked every field	No per-record sign-off in batch mode — batch export is an analyst path, not a clinical one
Transmission inspector	WORKING	The exact endpoint, model, full prompt and de-identified payload that left the browser — and the identifier map that did not	Same view for batch runs
PEPFAR dataset mode — the engine this was adapted from	WORKING	Column auto-mapping, MER indicators that each carry their reason, DQ rules with per-facility scoring	Outside this track's scope
EMR / FHIR export	PROPOSED	CSV download exists today	FHIR resource mapping — not started
OCR / photo capture of handwritten notes	PROPOSED	Nothing in the code accepts an image	A new input stage ahead of de-identification — not built

Disclosure. The extraction, de-identification and rules engine were originally built for an HIV programme reporting use case and adapted to this track's dataset today. The clinical schema profile, the documentation quality rules and the evaluation harness were built during the event. Gemini 2.5 Flash is a third-party foundation model accessed via API — it is not our model.

Stack. React 19, TypeScript, Tailwind 4, Vite, PapaParse, SheetJS, lucide-react and the Anthropic SDK — all open source. No charting library; the bars are CSS.

Models. No model was trained or fine-tuned. The only AI component is a hosted foundation model called over HTTPS. Two providers are wired up in `extract.ts` and chosen at runtime: Google `gemini-2.5-flash` (default) and Anthropic `claude-sonnet-4-6`. The key is entered in the page, kept in memory, and sent only to the provider you pick.

CLAIM

MER Capture extracts structured clinical fields and ICD-10 codes from free-text consultation notes at a **factual-match rate high enough to make clinician review faster than manual documentation**, without inventing facts absent from the source.

TEST POPULATION & SETTING

Consultation narratives from the organiser dataset. Deployment target: Nigerian secondary and tertiary outpatient clinics.

GROUND TRUTH

The reference structured fields paired with each narrative. Extraction sees the narrative alone; only fields with a matching reference column are scored.

HELD-OUT SET

{{TEST_SET_SIZE}} records — a seeded shuffle, default 20%, with the seed and set size shown on screen so the split is reproducible and stated. The prompt is fixed in `extract.ts` and holds no dataset-specific examples.

RESULTS ON THE HELD-OUT SET

EXACT MATCH

{{EXACT_MATCH_PCT}}

TOKEN F1

{{FUZZY_F1}}

ICD-10 EXACT

{{ICD10_EXACT}}

ICD-10 CATEGORY

{{ICD10_CATEGORY}}

HALLUCINATION RATE

{{HALLUCINATION_RATE}}

NEGATION ERRORS

{{NEGATION_ERRORS}}

Definitions, as implemented. *Exact* — equality after lowercasing, collapsing whitespace and stripping trailing punctuation; macro-averaged across fields, micro also reported. *Token F1* — bag-of-words F1 with stopwords removed, so correct content phrased differently still scores. *ICD-10* — full code, and separately the 3-character category (J45.9 counts against J45.0). *Hallucination* — field populated where the reference is empty, over the count of empty references. *Omission* — field empty where the reference is not, over the count of scored pairs. *Negation errors* — term-level polarity inversions: a term the reference denies that the extraction asserts, or the reverse. Every number carries its denominator on screen.

SUCCESS THRESHOLD

≥85% field-level factual match and <2% hallucination rate would justify a supervised clinical pilot.

FAILURE CRITERIA

Any hallucination rate above 5%, or any negation inversion on a safety-relevant field — allergy, medication, symptom denial — stops deployment pending redesign.

BASELINE COMPARISON

Not measured. We have not yet run regex / keyword section-splitting on the same held-out set, so we cannot claim extraction beats the simplest non-AI alternative. That baseline is the next thing we build.

FAIRNESS

Performance should be checked separately across age, sex and presenting condition. Whether the dataset permits this depends on those columns being populated — where they are absent, the subgroup result is reported as unavailable rather than assumed equal.

What it takes to run this in a clinic, and what happens when it is wrong.

DEPLOYMENT CONTEXT

Browser-based. No server of ours, no GPU. Runs on an ordinary facility laptop. De-identification happens on the device — but the call to the model provider still needs the internet, so this is not an offline tool.

BIGGEST ASSUMPTION

That reviewing a pre-filled structured note is meaningfully faster than typing it — and that clinicians will actually correct low-confidence fields rather than rubber-stamp them.

BIGGEST RISK

Automation bias. A confident, fluent, wrong extraction being confirmed without scrutiny.

SAFEGUARDS — ALL OF THESE ARE IN THE CODE

- Confidence score on every extracted field.
- Source span on every field, located back in the note — hover a field and the sentence highlights in the original.
- Named sign-off gate: in the single-note view, export stays disabled until a clinician types their name and ticks the review box. **Batch export is not yet gated.**
- Absence is a valid answer — the model is told to omit rather than guess, and empty values are dropped before they reach a field.
- Deterministic rules for every derived flag and metric — no number on screen is model-generated.
- De-identification on device before any call, and an inspector showing the exact prompt and payload that left the browser alongside the identifier map that did not.
- No storage of any kind — no `localStorage`, no server, nothing survives the tab.

WHEN IT IS WRONG

- The source span makes the error visible at the point of review.
- Low confidence is flagged before the clinician signs.
- The clinician overrides.
- The confirmed record — not the extraction — is what leaves the system.

OUR AI MUST NEVER

Generate a diagnosis, a code, or a clinical fact that is not present in the source note — or export any record without named clinician confirmation.

Enforced today in the single-note view. Batch mode exports without a per-record gate — closing that is the first thing on the list, and we would not run a pilot until it is closed.

NEXT 30 DAYS

A time-and-accuracy study: **5 clinicians × 20 encounters each**, half documented manually and half by reviewing MER Capture output. Measuring documentation time, field completeness, coding accuracy and clinician-corrected error rate.